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SPECT Quantitative Analysis Assessment

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Summary
SPECT imaging was acquired at PPMI imaging centers per the PPMI imaging protocol and sent to the imaging core lab for quality control review, reconstruction, and visual interpretation. Quantitative analysis was performed on all SPECT imaging by XingImaging utilizing the methodologies outlined in this document.

Method
Quantitative analysis of DAT SPECT imaging for determination of dopamine transporter specific binding levels was performed at XingImaging by expert image analysts in the field of nuclear medicine. All analyses were performed using the quantitative image analysis software MIAKAT (version 5.0) to derive the Specific Binding Ratio (SBR) values as the outcome measure of interest for a set of relevant target regions (see Table 1). Two sets of analysis outcome measures were generated using both the cerebral white matter (CWM) and occipital cortex (OCC) as reference regions.

Level	**Total Region**	**Right Region**	**Left Region**
1	Striatum	Striatum_R	Striatum_L
2	Caudate	Caudate_R	Caudate_L
2	Putamen	Putamen_R	Putamen_L
3	PreCaudate	PreCaudate_R	PreCaudate_L
3	PosCaudate	PosCaudate_R	PosCaudate_L
3	Precommissural_Putamen	Precommissural_Putamen_R	Precommissural_Putamen_L
3	Poscommissural_Putamen	Poscommissural_Putamen_R	Poscommissural_Putamen_L
4	PreDorsalPutamen	PreDorsalPutamen_R	PreDorsalPutamen_L
4	PreVentralPutamen	PreVentralPutamen_R	PreVentralPutamen_L
4	PosDorsalPutamen	PosDorsalPutamen_R	PosDorsalPutamen_L
4	PosVentralPutamen	PosVentralPutamen_R	PosVentralPutamen_L

Table 1: List of target regions

The input images used in the processing of these data were;
- 123I-DaTscan™ SPECT data: derived from a standard reconstruction on a HERMES workstation

All DAT SPECT images were processed in MIAKAT as follows;

- Spatial normalization of DAT SPECT data into stereotaxic space (MNI152):

  - The DAT SPECT data were mapped directly into MNI152 space using an affine transformation to a standard DAT_template using a 12-parameter affine registration
  - Resulting DAT SPECT images in MNI152 space were visually assessed to confirm successful mapping into stereotaxic space.
    - If spatial normalization was *successful*, analysis proceeded to the next phase.
    - If spatial normalization was *unsuccessful*, alternative algorithm parameters or manual perturbation of parameters were investigated to see if an appropriate spatial normalization could be achieved.
      - If spatial normalization was *successful*, analysis proceeded to the next phase.
      - If spatial normalization was *unsuccessful*, subsequent analysis was not performed for this scan.

- Application of target and reference regions of interest using the CIC atlas:
  - Target regions from the CIC atlas (listed in Table 1) were applied to the DAT SPECT data in MNI152 space to calculate the average image value for each region (SUV^{TAR}).
  - The reference region, defined as either the cerebral white matter or occipital cortex in the CIC atlas, was applied to the DAT SPECT data in MNI152 space to calculate the average image value for the reference region (SUV^{REF}).

- Calculation of regional SBR outcome measures:
  - The SBR values for the set of target regions of interest are calculated as;

$$SBR^{TAR} = (SUV^{TAR} / SUV^{REF}) - 1$$

These calculations were performed for both the cerebral white matter and occipital cortex reference regions.